RAJALAKSHMI ENGINEERING COLLEGE

ABINAVVA 251001504-2025-IT

A2

Started on	Tuesday, 22 September 2026, 8:47 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 8:49 AM
Time taken	2 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    count++;                // n == 1
    if(n == 1)
    {
        count++;            // printf("*")
    }
    else
    {
        for(int i = 1; ; i++)
        {
            count++;        // i <= n
            if(i > n)
                break;

            for(int j = 1; ; j++)
            {
                count++;    // j <= n
                if(j > n)
                    break;

```

```
                count++;      // first printf("*")
                count++;      // second printf("*")

                break;
            }

            count++;          // i++
        }
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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